

$$1. \quad E = 1 + 4u^2 \quad F = -4uv \quad G = 1 + 4v^2$$

$$L = \frac{2}{\sqrt{1+4u^2+4v^2}} \quad M = 0 \quad N = \frac{-2}{\sqrt{1+4u^2+4v^2}}$$

Bonnet 定理

$$E, G > 0 \quad \boxed{EG - F^2 > 0} \quad \boxed{\text{Gauss.}} \quad \boxed{\text{Peterson-Codazzi-Mainardi}}$$

$$EG - F^2 = (1 + 4u^2)(1 + 4v^2) - (-4uv)^2 = 1 + 4u^2 + 4v^2 > 0$$

Gauss 方程

$$EK = (\underline{r_{11}^2})_v - (\underline{r_{12}^2})_u + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - \Gamma_{12}^2 \Gamma_{12}^2$$

$$\underline{FK} = (\underline{r_{12}^2})_u - (\underline{r_{11}^2})_v + \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1$$

$$GK = (\underline{r_{22}^2})_u - (\underline{r_{12}^2})_v + \Gamma_{22}^1 \Gamma_{11}^1 + \Gamma_{22}^2 \Gamma_{12}^1 - \Gamma_{12}^1 \Gamma_{12}^1 - \Gamma_{12}^2 \Gamma_{22}^2$$

$$K = \frac{\det(II)}{\det(I)} = \frac{LN - M^2}{EG - F^2} = \frac{-4}{1 + 4u^2 + 4v^2} = \frac{-4}{(1 + 4u^2 + 4v^2)^2}$$

$$Eu = 8u \quad Fu = -4v \quad Gu = 0$$

$$Ev = 0 \quad Fv = -4u \quad Gv = 8v$$

$$\Gamma_{11}^1 = \frac{GEu - 2FFu + FEv}{2(EG - F^2)}$$

$$= \frac{(1 + 4v^2) \cdot 8u - 2 \cdot (-4uv) \cdot (-4v) + (-4uv) \cdot 0}{2(1 + 4u^2 + 4v^2)}$$

$$= \frac{4u}{1 + 4u^2 + 4v^2}$$

$$\Gamma_{11}^2 = \frac{2EFu - EEv - FEu}{2(EG - F^2)}$$

$$= \frac{2(1 + 4u^2) \cdot (-4v) - (1 + 4u^2) \cdot 0 - (-4uv) \cdot 8u}{-4v \cdot 2(EG - F^2)}$$

$$= \frac{-4v}{1 + 4u^2 + 4v^2}$$

$$\begin{aligned} \Gamma_{12}^1 &= \frac{G E v - F G u}{2(EG - F^2)} \\ &= \frac{(1+4v^2) \cdot 0 - (-4uv) \cdot 0}{2(1+4u^2+4v^2)} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \Gamma_{12}^2 &= \frac{E G u - F E v}{2(EG - F^2)} \\ &= \frac{(1+4u^2) \cdot 0 - (-4uv) \cdot 0}{2(1+4u^2+4v^2)} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \Gamma_{22}^1 &= \frac{2 G F v - G G u - F G v}{2(EG - F^2)} \\ &= \frac{2(1+4v^2) \cdot (-4u) - (1+4v^2) \cdot 0 - (-4uv) \cdot (8v)}{2(1+4u^2+4v^2)} \\ &= \frac{-4u}{1+4u^2+4v^2} \end{aligned}$$

$$\begin{aligned} \Gamma_{22}^2 &= \frac{E G v - 2 F F v + F G u}{2(EG - F^2)} \\ &= \frac{(1+4u^2)(8v) - 2(-4uv)(-4u) + (-4uv) \cdot 0}{2(1+4u^2+4v^2)} \\ &= \frac{4v}{1+4u^2+4v^2} \end{aligned}$$

$$\begin{aligned} (\Gamma_{11}^1)_v &= \left(\frac{4u}{1+4u^2+4v^2} \right)_v \\ &= \frac{-4u \cdot 8v}{(1+4u^2+4v^2)^2} \\ &= \frac{-32uv}{(1+4u^2+4v^2)^2} \end{aligned}$$

$$(\Gamma_{12}^1)_u = 0 \quad (\Gamma_{12}^1)_v = 0$$

$$(T_{12}^1)_u = \left(\frac{-4u}{1+4u^2+4v^2} \right)_u$$

$$= \frac{-4(1+4u^2+4v^2) - (-4u) \cdot 8u}{(1+4u^2+4v^2)^2}$$

$$= \frac{-4 - 16u^2 - 16v^2 + 32u^2}{(1+4u^2+4v^2)^2}$$

$$= \frac{-4 + 16u^2 - 16v^2}{(1+4u^2+4v^2)^2}$$

$$(T_{11}^2)_v = \left(\frac{-4v}{1+4u^2+4v^2} \right)_v$$

$$= \frac{-4(1+4u^2+4v^2) - (-4v)(8v)}{(1+4u^2+4v^2)^2}$$

$$= \frac{-4 - 16u^2 + 16v^2}{(1+4u^2+4v^2)^2}$$

$$(T_{12}^2)_u = 0$$

Peterson - Codazzi - Mainardi 方程

$$L_v - M_u = L T_{12}^1 + M(T_{12}^2 - T_{11}^1) - N T_{11}^2$$

$$M_v - N_u = L T_{22}^1 + M(T_{22}^2 - T_{12}^1) - N T_{12}^2$$

$$L = \frac{2}{\sqrt{1+4u^2+4v^2}}$$

$$L_v = \frac{-2 \cdot \frac{1}{2} (1+4u^2+4v^2)^{-\frac{1}{2}} \cdot 8v}{1+4u^2+4v^2} = \frac{-8v}{(1+4u^2+4v^2)^{\frac{3}{2}}}$$

$$M = 0 \quad M_u = M_v = 0$$

$$N = \frac{-2}{\sqrt{1+4u^2+4v^2}}$$

$$N_u = \frac{2 \cdot \frac{1}{2} (1+4u^2+4v^2)^{-\frac{1}{2}} \cdot 8u}{1+4u^2+4v^2} = \frac{8u}{(1+4u^2+4v^2)^{\frac{3}{2}}}$$

$$r(u, v) = (u, v, u^2 - v^2)$$

2. 由

$$4(EG - F^2)^2 K$$

$$= E \left(\frac{\partial E}{\partial q} \frac{\partial G}{\partial q} - 2 \frac{\partial F}{\partial p} \frac{\partial G}{\partial q} + \left(\frac{\partial G}{\partial p} \right)^2 \right)$$

$$+ F \left(\frac{\partial E}{\partial p} \frac{\partial G}{\partial q} - \frac{\partial E}{\partial q} \frac{\partial G}{\partial p} - 2 \frac{\partial E}{\partial q} \frac{\partial F}{\partial q} + 4 \frac{\partial E}{\partial p} \frac{\partial F}{\partial q} - 2 \frac{\partial F}{\partial p} \frac{\partial G}{\partial p} \right)$$

$$+ G \left(\frac{\partial E}{\partial p} \frac{\partial G}{\partial p} - 2 \frac{\partial E}{\partial p} \frac{\partial F}{\partial q} + \left(\frac{\partial E}{\partial q} \right)^2 \right)$$

$$- 2(EG - F^2) \left(\frac{\partial^2 E}{\partial q^2} - 2 \frac{\partial^2 F}{\partial p \partial q} + \frac{\partial^2 G}{\partial p^2} \right)$$

$$\text{则 } K = \frac{u^2 v^2 (3u^4 + 2u^3 v + 2u v^3 + 3v^4) - 2u^2 v^2}{(u^2 v^2 + u^2 v^2)^2}$$

3. 证明有十个洞的甜甜圈面不是直纹曲面

$$K \leq 0$$

存在椭圆点 $K > 0$

不是直纹曲面.